The Grushin Bochner–Riesz conjecture in arXiv:1110.3227

Answered, with a stronger sharp threshold, by arXiv:1204.1159

fa_banach_001, lane 4

13 August 2026

Status: literature already answered. Theorem 2 of Martini–Sikora proves the source's conjectured uniform L^p boundedness and improves its requested exponent from $\delta > (n+1)/2$ to the sharp range $\delta > n/2$.

1. Exact source conjecture

Jotsaroop, Sanjay, and Thangavelu study

$$G = -\Delta_x - |x|^2 \partial_t^2 \quad \text{on } \mathbb{R}_x^n \times \mathbb{R}_t$$

and its Bochner–Riesz means

$$B_R^\delta = (1 - G/R)_+^\delta.$$

Their Theorem 1.4 establishes uniform L^p boundedness for $\delta > (n+1)/2 + 1/6$, $1 < p < \infty$. They then conjecture the improved range

$$\delta > \frac{n+1}{2}, \qquad n \geq 2. \tag{1}$$

Here $\left(1 - \frac{H(\lambda)}{R}\right)_+^\delta f^\lambda$ are the Bochner-Riesz means associated to $H(\lambda)$. For λ fixed these means have been studied by various authors, see [**9**]. Concerning $B_R^\delta f$ for the Grushin operator we prove the following result.

Theorem 1.4. *For $\delta > (n+1)/2 + 1/6$, the Bochner-Riesz means B_R^δ are uniformly bounded on $L^p(\mathbb{R}^{n+1}), 1 < p < \infty$.*

It may not be possible to improve the above result when $n = 1$ as the critical index for the Bochner-Riesz summability of one dimensional Hermite expansions is $1/6$, see [**9**]. However, the critical index for Hermite expansions on $\mathbb{R}^n, n \geq 2$ is $(n-1)/2$ and hence it should be possible to improve the above result. We conjecture that the above result is true for $\delta > (n+1)/2$ for $n \geq 2$.

Figure 1: Theorem 1.4 and the explicit conjecture on page 3 of arXiv:1110.3227.

2. Exact later theorem

Martini and Sikora consider the more general Grushin operator

$$L = -\Delta_{x'} - |x'|^2 \Delta_{x''} \quad \text{on } X = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$$

and define

$$D = \max\{d_1 + d_2, 2d_2\}.$$

Their Theorem 2 states that, for every $p \in [1, \infty]$ and

$$\kappa > \frac{D-1}{2}, \tag{2}$$

the means $(1 - tL)_+^\kappa$ are bounded on $L^p(X)$ uniformly in $t \geq 0$.

where $F_{(t)}(\lambda) = F(t\lambda)$, and $\eta \in C_c^\infty(]0,\infty[)$ is a not identically zero auxiliary function; note that different choices of η give rise to equivalent local norms. Next set $D = \max\{d_1 + d_2, 2d_2\}$. Then our main result reads as follows.

Theorem 2. *Suppose that $\kappa > (D-1)/2$ and $p \in [1, \infty]$. Then the Bochner-Riesz means $(1 - tL)_+^\kappa$ are bounded on $L^p(\mathrm{X})$ uniformly in $t \in [0, \infty[$.*

As in many other works on the subject, the proof of our results is based on the analysis of the integral kernel $\mathcal{K}_{F(L)} : \mathrm{X} \times \mathrm{X} \to \mathbb{C}$ of the operator $F(L)$, defined by

Figure 2: Two cropped excerpts: the definition of D and Theorem 2 on page 2 of arXiv:1204.1159.

3. Parameter identification and implication

The source operator is exactly the later operator with

$$d_1 = n, \qquad d_2 = 1, \qquad x' = x, \qquad x'' = t.$$

Thus, for every $n \geq 1$,

$$D = \max\{n+1, 2\} = n+1.$$

Substitution into (2) gives

$$\delta = \kappa > \frac{D-1}{2} = \frac{n}{2}.$$

The scale parameters agree after putting $t = R^{-1}$: $(1 - tL)_+^\kappa = (1 - G/R)_+^\delta = B_R^\delta$. Consequently Martini–Sikora prove uniform boundedness on every $L^p(\mathbb{R}^{n+1})$, including the source's range $1 < p < \infty$, whenever $\delta > n/2$. Since

$$\frac{n+1}{2} > \frac{n}{2},$$

this strictly implies the conjectured statement (1).

The supporting paper also states that its multiplier and Bochner–Riesz thresholds are sharp when $d_1 \geq d_2$. Here $n = d_1 \geq 1 = d_2$, so the later result not only answers the source conjecture but identifies the stronger sharp open-threshold range.

4. Provenance, search, and scope

The four lightweight run indexes contained no entry for either arXiv id. Bounded searches by the exact title, operator, and conjectured exponent found arXiv:1204.1159. Its introduction explicitly compares its Theorems 1 and 2 with the earlier $d_2 = 1$ work, and its bibliography identifies that work as Jotsaroop–Sanjay–Thangavelu. The formula-level specialization above removes any ambiguity caused by the supporting paper's nearby remark that Theorem 1 answers a different multiplier conjecture in the special case $d_1 = d_2 = 1$.

No new proof is claimed here. This packet records an exact primary-source literature resolution and the parameter conversion showing that Theorem 2 is strictly stronger than the queued conjecture.

References

[1] K. Jotsaroop, P. K. Sanjay, and S. Thangavelu, *Riesz transforms and multipliers for the Grushin operator*, arXiv:1110.3227 (2011); J. Analyse Math. 119 (2013), 255–273.

[2] A. Martini and A. Sikora, *Weighted Plancherel estimates and sharp spectral multipliers for the Grushin operators*, arXiv:1204.1159 (2012); Math. Res. Lett. 19 (2012), 1075–1088.